

Java Collections – Learning Guide

1. Collections Framework

The Java Collections Framework provides interfaces and classes for storing and manipulating groups of objects. Common interfaces include List, Set, Map, and Queue.

2. List

List is an ordered collection and generally permits duplicate elements. ArrayList is a commonly used List implementation.

Example: `ArrayList names = new ArrayList<>();`

3. Set

Set represents a collection that does not allow duplicate elements. HashSet is a common implementation.

4. Map

Map stores key-value pairs. Keys are unique within a Map. HashMap is a commonly used implementation.

Example: `HashMap students = new HashMap<>();`

5. Queue

Queue is designed for holding elements before processing. Common operations include adding elements and removing the next element.

6. ArrayList vs HashSet

ArrayList preserves element order and allows duplicates. HashSet does not allow duplicate elements and does not provide the same indexed access as ArrayList.

7. HashMap

HashMap stores mappings between keys and values. A value can be retrieved using its key. Example: `students.put(1, "Ravi");` and `students.get(1)`.

8. Quick Review

Use List when order and duplicates matter, Set when duplicate elements should be avoided, and Map when data is represented as key-value pairs.